
Time On-Site Analysis- E-Commerce Sector

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Overview

This analysis explores user engagement on an e-commerce site by examining time-on-site metrics across various dimensions, including device type, referral source, and page performance.

Objective

The primary goals were to analyze user time on the site, identify engagement trends, and assess the effectiveness of high- and low-engagement pages.

Assigned Task(s)

Time On-Site Analysis - E-Commerce Sector

Task Details

- Task 44: Conducted a detailed time on-site analysis, including A/B testing to compare user behavior across different engagement levels.
- Status: Completed
- Details:
 - Data Preparation: Cleaned and pre-processed the dataset to ensure accuracy in the analysis.
 - Time Spent Analysis: Calculated average time spent by device type, referral source, and geographic location to identify user engagement trends.
 - Bounce Rate Assessment: Analyzed bounce rates across device types to understand user exit patterns.
 - Session Duration Distribution: Created a histogram to visualize session duration across various intervals, revealing user behavior patterns.
 - High Engagement Page Analysis: Identified core pages with the highest total time spent, indicating key user interaction points.
 - Heatmap Creation: Developed a heatmap to visualize user activity by day of the week and hour, highlighting peak engagement times.
 - A/B Testing: Conducted tests comparing high and low engagement pages to evaluate differences in average time spent and conversion rates.
 - Insights Generation: Compiled findings into actionable insights for improving user engagement and optimizing low-engagement pages.

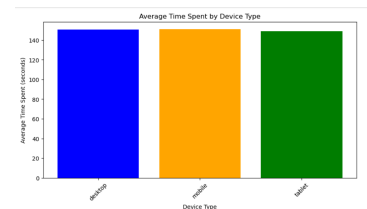
Progress

- Accomplishments:
 - Comprehensive Analysis Completed: Successfully performed a detailed analysis of user engagement metrics, providing insights into user behavior across multiple dimensions.
 - Engagement Metrics Identified: Established baseline metrics for average time spent by device type, referral source, and geographic location, facilitating targeted optimization strategies.
 - Key Insights Gained: Discovered that average time spent was similar across devices, while notable differences in engagement were observed based on referral sources and specific pages.
 - A/B Testing Results: Completed A/B testing that revealed significant differences in engagement levels between high and low engagement pages, informing future optimization efforts.
 - Actionable Recommendations Developed: Generated actionable insights from the analysis, leading to potential improvements in user experience and engagement on low-performing pages.
 - Data Visualization Created: Developed visual representations of user activity and engagement patterns, enhancing the understanding of user interactions on the site.

- Metrics:

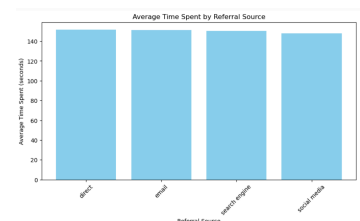
- Average Time Spent by Device Type:

- Desktop: 150.51 seconds
 - Mobile: 151.09 seconds
 - Tablet: 149.17 seconds



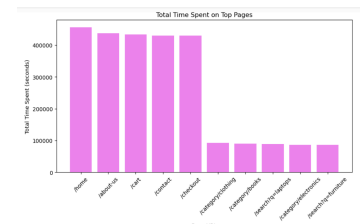
- Average Time Spent by Referral Source:

- Direct: 151.56 seconds
 - Email: 151.10 seconds
 - Search Engine: 150.37 seconds
 - Social Media: 148.02 seconds

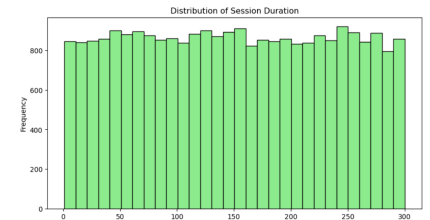


- Average Time Spent on Pages:

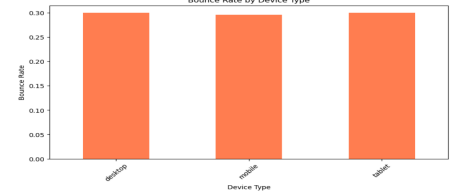
- Highest Engagement Page: /home – 400,000+ seconds total time spent
 - Lower Engagement Pages: Category pages (e.g., /category/clothing) – approximately 100,000 seconds or less.



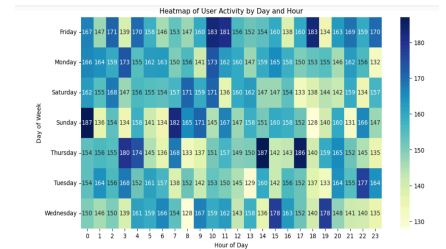
- Session Duration Distribution: Histogram indicating a uniform distribution of session durations across various intervals (0 to 300 seconds).



- Bounce Rates:
 - Desktop: 30%
 - Mobile: 30%
 - Tablet: 30%



- Heatmap Insights: Peak user activity was observed late at night and early morning hours, especially on weekends.



- A/B Testing Results:
 - Average Time Spent (Control Group - High Engagement): 150.71 seconds
 - Average Time Spent (Test Group - Low Engagement): 17.90 seconds
 - Conversion Rate (Control Group): 9.76%
 - Conversion Rate (Test Group): 10.71%

Challenges and Solutions

- Challenge Faced:
 - Data Inconsistencies: Encountered missing or inconsistent values in critical columns (e.g., *Event Duration*), affecting the accuracy of engagement calculations.
 - Low Engagement Sample Size: Low engagement pages had limited session data, resulting in small sample sizes for A/B testing and skewed averages.
 - Complexity in Identifying High vs. Low Engagement Pages: Defining high and low engagement thresholds required multiple adjustments to ensure accurate segmentation.
 - Time on Site Calculation Complexity: Deriving time on site from individual events demanded extra processing steps and cross-referencing timestamps.
- Solutions Implemented:
 - Data Cleaning: Applied data-cleaning techniques, such as filling missing values with median values and filtering incomplete records, to maintain dataset integrity.

- Sample Size Adjustments: Expanded page selection criteria for low engagement pages to increase sample size for A/B testing, yielding more reliable averages.
- Segmentation Refinement: Iteratively adjusted the engagement threshold for pages to identify high and low engagement categories more accurately.
- Enhanced Calculation Logic: Refined the *Time Spent* calculations by focusing on session timestamps and leveraging time-based aggregates, resulting in more precise metrics.

Next Step

- Upcoming Task: Plan to explore further optimizations for low-engagement pages and test additional variants to improve user engagement.
- Goals: Set a target to increase the average time spent on low-engagement pages by at least 10% through design and content improvements.

Conclusion

- Summary: In this time on-site analysis, we conducted a comprehensive examination of user engagement metrics across device types, referral sources, and specific pages on the site. Key findings revealed consistent engagement across devices and referral sources, with desktop and mobile users slightly more engaged than tablet users. High-engagement pages, such as the homepage and checkout page, showed the most time spent, while category pages demonstrated lower engagement. A heatmap provided insights into peak user activity during late-night and early-morning hours, especially on weekends. A/B testing between high and low-engagement pages further highlighted significant differences in user engagement and conversion rates, supporting recommendations for optimization on low-engagement pages. Despite challenges with data inconsistencies and sample size limitations, we successfully implemented data-cleaning and segmentation refinements, resulting in reliable metrics and actionable insights. This analysis has set a strong foundation for targeted improvements in user experience, ultimately aimed at increasing site engagement and conversions.
- Acknowledgment: Thank you for your time and attention.